

CHE 321 - Exam 1 - Sample 3

examfun.m

```
1  function h=examfun(~,y)
2
3  x = y(1);
4  p = y(2);
5
6  cao = 2;
7  epi = 3;
8  k = 10;
9  kc = 200;
10 z = 0.07;
11 dp = 1;
12 alpha = z/dp^2;
13 fao = 10;
14 %i know alpha is inversly proportional to Dp^2 so (alpha = z/Dp^2) for some
15 %constant z, and i also have the value for alpha as 0.07 at Dp = 1mm, so for changing
16 %i can get the value of z = 0.07
17
18 ca = (cao * (1 - x) * p) / (1 + (epi * x));
19 cb = (cao * (x) * p) / (1 + (epi * x));
20 cc = (cao * (3*x) * p) / (1 + (epi * x));
21
22 r = -k * (ca - ((cb * cc^3) / kc));
23
24 h1 = (-alpha * (1 + (epi * x)))/ (2*p);
25 h2 = -r/fa0;
26
27 h = [h1,h2]';
```

```
1  clc;
2  clear all;
3
4  y0 = [0 1];
5  wspan = [1,10];
6
7  [w,y] = ode45(@examfun,wspan,y0);
8
9  figure(1);
10 plot(w,y(:,1));
11 xlabel('W');
12 ylabel('conversion');
13 title('W vsConcentration');
14 grid on;
15
16 figure(2);
17 plot(w,y(:,2));
18 xlabel('W');
19 ylabel('pressure');
20 title('W vs P');
21 grid on;
22
23 % now for part 3 i change the Dp values from 0.01 to 10 and create a plot
24 % against conversion, calling Dp from the previous function and changing its
25 % values using a for loop in the range (0.01 to 10) with an increment of 0.5 recalculating the defrential equations by calling
26 % the ODE function again, then i can plot those values using the plot and
27 % hold function
28 %@examfund{dp}= [5];
```